10 critical metrics used by companies to measure RAG performance

In production **Retrieval-Augmented Generation (RAG)** systems, companies measure several metrics to ensure **retrieval quality, answer correctness, latency, and reliability**. Tools like RAGAS, Arize Phoenix, and LangSmith commonly track these metrics.

Below are the **10 most critical metrics used in enterprise RAG systems**.

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1 Context Precision

What it measures

How many of the retrieved documents are actually relevant to the query.

Formula concept:

Relevant Retrieved Docs / Total Retrieved Docs

Example:

Retrieved Docs	Relevant
5	4

Score:

Precision = 4 / 5 = 0.80

High precision means **the retriever is not returning irrelevant documents**.

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2 Context Recall

What it measures

Whether the retriever fetched **all important documents needed to answer the query**.

Example:

Relevant Docs in DB	Retrieved
5	3

Score:

Recall = 3 / 5 = 0.60

Low recall means **important information was missed**.

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3 Faithfulness (Groundedness)

What it measures

Whether the answer is **supported by the retrieved context**.

Example:

Context:

```
Waiting period: 30 days
```

Answer:

```
Waiting period is 60 days
```

This is **not faithful**.

Faithfulness checks if statements **exist in the context**.

---

4 Answer Relevance

What it measures

How well the generated answer **addresses the user’s question**.

Example:

Question:

```
What is the waiting period?
```

Answer:

```
The policy offers hospitalization coverage.
```

This answer is **not relevant**, even if it came from context.

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5 Hallucination Rate

What it measures

Percentage of responses containing **information not supported by context**.

Formula:

Hallucination Rate = Hallucinated Responses / Total Responses

Example:

Responses	Hallucinated
100	7

Score:

7%

Companies try to keep hallucination **below 5%**.

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6 Retrieval Latency

What it measures

Time required for the retriever to fetch documents from the vector database.

Example pipeline:

```
User Query
   ↓
Embedding
   ↓
Vector Search
```

Typical values:

System	Latency
Good RAG	<100 ms
Slow RAG	>500 ms

Vector databases like Qdrant are optimized for low latency retrieval.

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7 LLM Response Latency

What it measures

Total time the LLM takes to generate an answer.

Example:

Model	Latency
Small model	1–2 sec
Large model	4–10 sec

Companies monitor this for **user experience optimization**.

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8 Token Usage

What it measures

Number of tokens consumed per request.

Includes:

- prompt tokens
- context tokens
- completion tokens

Example:

```
Prompt tokens: 600
Completion tokens: 150
Total: 750
```

This metric affects **cost and speed**.

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9 Retrieval Accuracy

What it measures

How often the retriever fetches **documents containing the correct answer**.

Example:

Queries	Correct Context Retrieved
100	84

Score:

84%

Low retrieval accuracy means:

- poor embeddings
- bad chunking
- incorrect retriever parameters

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10 User Satisfaction Score

What it measures

How satisfied users are with the generated answers.

Methods used:

- thumbs up/down feedback
- ratings (1–5)
- user surveys

Example:

Feedback	Percentage
Positive	92%
Negative	8%

This is the **ultimate business metric**.

📊 How These Metrics Fit in the RAG Pipeline

User Query
↓
Retriever
↓
Context
↓
LLM
↓
Answer

Evaluation layers:

Stage	Metrics
Retriever	Context Precision, Context Recall
Generation	Faithfulness, Answer Relevance
Quality	Hallucination Rate
Performance	Latency, Token Usage
Business	User Satisfaction

🏗 Enterprise RAG Monitoring Stack

Typical monitoring architecture:

```
RAG Application
   ↓
Observability Layer
   ↓
Metrics Dashboard
```

Monitoring tools include:

- Arize Phoenix
- LangSmith
- RAGAS

These platforms show:

- query traces
- hallucination detection
- RAG performance metrics

📌 Simple Way to Remember

RAG performance depends on **three pillars**:

```
Retrieval Quality
Generation Quality
System Performance
```

Metrics:

```
Retrieval → Precision, Recall, Accuracy
Generation → Faithfulness, Relevance, Hallucination
Performance → Latency, Tokens
Business → User Satisfaction
```